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Perceptions of the educational environment during clinical training among Vietnamese medical students: a cross-sectional study

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Abstract

Background Gathering feedback from medical students about their learning environment plays a vital role in improving the quality of medical education. Moreover, no studies in Vietnam have examined the diversity of factors such as mental health and perceived quality of clinical training in relation to perceptions of the educational environment.

Aim We aimed to examine Vietnamese medical doctor students' perceptions of the educational environment and its associated factors.

Methods We conducted a cross-sectional study using multistage cluster random sampling technique to recruit medical students who were in their clinical training years (4th, 5th, and 6th year) at Hanoi Medical University in Vietnam from November 2015 to January 2016. Students self-reported their demographic characteristics, academic performance, mental health, academic motivation, and perceptions of the educational environment (via The Dundee Ready Education Environment Measure (DREEM)). The modified Poisson regression models were used to assess the association between potential risk factors and negative perceptions of educational environment (i.e., score ≤ 100/200).

Results Among 484 students enrolled in the study (response rate 77.2%), 87.8% reported positive perception (DREEM score > 100) of medical education environment, with the highest satisfaction for teachers. Senior year status was significantly associated with the negative perception in both sexes, while having self-reported mental health issues (i.e., depression and stress) and no academic motivation were associated with negative perceptions among male students only. Additionally, fulfillment of learning objectives, perceived effectiveness and assessment fairness of clinical training were also important associated factors of students' positive perception of the educational environment.

Conclusions The majority of medical students in this study expressed overall satisfaction with the learning environment, particularly with their teachers. However, the findings underscore the need for targeted improvements in learning effectiveness, fair assessment, especially for students in senior years, those facing mental health challenges or lacking academic motivation.

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Keywords Educational environment, Dundee Ready Education Environment Measure (DREEM), Clinical training, Medical students, Perception, Vietnam

Background

According to the World Federation for Medical Education, the educational environment as a key focus for evaluating medical education programs [1]. Given that the importance of the educational environment has been widely acknowledged [2, 3], Vietnamese medical education institutions and schools often put a collective effort into creating a supportive learning environment by improving student welfare, teaching quality, curriculum design, offering further learning opportunities, and available facilities [4–9]. Therefore, it is vital to collect the feedback of medical students to help educators identify strengths and weaknesses that need to be addressed for quality improvement [10, 11].

Among medical training components, clinical practice is critical as it allows students to interact with real-life patients, engage in hands-on practice, and apply theoretical knowledge in real-world situations to develop clinical competencies [12]. In Vietnam, the medical curriculum is structured into two key phases, including basic sciences during the first three years and clinical sciences and rotations in the final three years, which aligns with competence standards for general practitioners issued by the Vietnam Ministry of Health [13]. In many countries, students' perceptions of the educational environment in medical school have been well studied [14–19]. In Vietnam, studies measuring the educational environment have been confined to nursing students, consistently reporting perception as "more positive than negative" (total scores of at least 120/200) [20–23], but there is no comparable research involving medical doctor students. Moreover, existing DREEM-based research globally has rarely examined the joint contributions of mental health issues (e.g., stress, depression, anxiety), academic motivation, and perceived clinical training quality, in relation to perceptions of the educational environment, or explored whether these association differ by sex. Therefore, this study aimed to examine medical students' perceptions of the educational environment during clinical practices and identify its associated factors among medical doctor students at Hanoi Medical University, the largest and most well-established medical school in Vietnam. The findings hope to provide a valuable foundation for improving clinical education quality, supporting evidence-based curricular reforms, and better preparing students for medical practices.

Methods

Study design and participants

A cross-sectional study was conducted among medical students studying the general practitioner track who were in their clinical training years (4th, 5th, and 6th year) at Hanoi Medical University between November 2015 and January 2016. Only clinical-year students (4th, 5th, and 6th year) were included, as the study aimed to assess perceptions during clinical training, including hospital-based rotations and bedside teaching, which preclinical students (1st–3rd year) have not experienced.

Using a multistage cluster sampling technique, four out of six classes were randomly selected from each cohort, followed by the random selection of two groups within each class. The primary sampling unit was groups of students, each of which contained 20–25 students. Finally, we had a sampling frame of 627 students in the selected groups. Eligible participants were invited via face-to-face interactions at various locations, including the university, hospitals, and dormitories. Upon completing the questionnaire, students were offered the opportunity to enter a lottery for one of three Littmann stethoscopes as a token incentive. The details of the method of this study have been described elsewhere [24].

Educational environment measure

The Dundee Ready Education Environment Measure (DREEM) was used to assess students' perceptions of the educational environment [25]. This 50-item inventory is divided into five domains: (1) students' perceptions of teaching (12 items, range: 0–48), (2) students' perceptions of teachers (11 items, range: 0–44), (3) students' academic self-perceptions (8 items, range: 0–32), (4) students' perceptions of atmosphere (12 items, range: 0–48), and (5) students' social self-perceptions (7 items, range: 0–28). Responses are rated on a 5-point Likert scale (4- "strongly agree", 3- "agree", 2- "unsure", 1- "disagree", 0- "strongly disagree"), with reverse scoring applied to nine negative items (4, 8, 9, 17, 25, 35, 39, 48, and 50). The total score is 200, and the results are interpreted using McAleer and Roff's guidelines: [25] 0–50 (very poor), 51–100 (plenty of problems), 101–150 (more positive than negative), and 151–200 (excellent). The questionnaire has been validated in multiple countries [19], but not in Vietnam at the time of our data collection. Therefore, in this study, we performed translation and a back-translation process involving three English-proficient physicians. There was no missing data observed for 50 DREEM items in the study sample. Moreover, we evaluated the internal consistency and underlying

factor structure of the DREEM using Cronbach's alpha coefficients. To ensure its readability and suitability for the local context, a pilot test of the questionnaire was conducted with a convenience sample of 52 medical students.

Covariates

The variables included demographic and academic factors: sex (male/female), academic year (4th, 5th, and 6th year), living arrangement (living with family, alone, or with roommates), level of English reading skills (basic, intermediate, proficient), whether students failed at least one clinical or theory exam (yes/no), grade point average (GPA) of the last academic year, academic ranking (excellent [8.0 and above], good [7.0—< 8.0], average [5.0—< 7.0], or weak [< 5.0]), the number of night shifts per week (< 2 times, or ≥ 2 times a week), and involvement in irrelevant-to-learning tasks during night shifts (yes or no); behaviors: currently smoking (yes or no) and regular alcohol consumption (> 1 time/month on average) (yes or no); mental health: screened positive for depression (yes or no), screened positive for generalized anxiety disorder (yes or no), stress at university (no—below median or yes -median and above); academic motivation (non-self-determined or self -determined); clinical training experiences: duration of clinical rotation (month), time spent at the hospital a day with/without scheduled clinical lecture to ask and examine patients (minutes), time spent at the hospital a day with/without scheduled clinical lecture to attend to clinician lectures (minutes), time spent at the hospital a day with/without scheduled clinical lecture to other tasks (e.g., reading a book, sitting in the hallway, coffee time, sleeping) (minutes), and number of medical records written per week; and perceived quality of clinical training: perceived that that all course objectives were completed (no, neutral, or yes), perceived that clinical learning was effective relative to time and effort invested (no, neutral, or yes), and perceived that clinical exam scores accurately reflected students' ability (no, neutral, or yes).

The validated Vietnamese version of Patient Health Questionnaire-9 (PHQ-9) was used to screen for depression [26]. Each question assessed how frequently participants had been bothered by mental health-related symptoms over the past two weeks, with response options including 0- "Not at all", 1- "Several days", 2- "More than half the days", 3- "Nearly every day". The total score was calculated across all 9 questions, and participants with scores of 10 or higher were categorized as screening positive for depression symptoms, following the approach used in previous study [24].

The Generalized Anxiety Disorder (GAD-7) was developed by Robert L. Spitzer et al [27] and has been used in some studies on the Vietnamese population [28, 29]. The questionnaire consists of 7 questions about the anxiety symptoms participant experienced during the past 2 weeks, with the response options including including 0- "Not at all", 1- "Several days", 2- "More than half the days", 3- "Nearly every day" based on the 4-point Likert scale. The total score was calculated by summing the answers to the 7 questions, ranging from 0 to 21. Participants with scores of 10 and higher were categorized as then would be defined as screened positive for generalized anxiety disorder.

The perceived medical school stress scale (PMSS) was used to assess students' perception about stress at university. This scale, developed by Dr. Peter Vitaliano et al., [30] consists of 13 questions with answers based on 5-point Likert scale, ranging from 0—"strongly disagree", 1- "disagree", 2- "neutral" to 3-"agree", 4- "strongly agree". The total score was calculated by summing all the answers for 13 questions, ranging from 0 to 52. In this study, besides being reported by overall score, the participants are divided into 2 groups, labeled as students having no stress when the PMSS score is lower than the median and the students with stress at school, with the PMSS score from the median and above.

The Academic Motivation Scale was used to assess self-determination motivation [31], following the approach used in previous study [24]. The scale comprises 28 statements assessing intrinsic motivation (to know, to accomplish tasks, and to experience stimulation), extrinsic motivation (external, introjected, and identified regulation), and amotivation. Participants rated their agreement with each statement on a 7-point Likert scale, including 1- "Does not correspond at all", 2 and 3- "Corresponds a little", 4- "Corresponds moderately", 5 and 6- "Corresponds a lot", 7- "Corresponds exactly". The total score was calculated using the formula: $[(2 \times \text{Intrinsic Motivation}) + (1 \times \text{Identified Regulation}) - (1 \times \text{External Regulation} + 1 \times \text{Introjected Regulation})/2 - (2 \times \text{Amotivation})]$. Based on the total score, participants were then categorized into two groups: self-determined academic motivation (i.e., score > 0, indicating a closer to intrinsic motivation) and non-self-determined academic motivation (i.e., score ≤ 0, indicating a closer to amotivation).

The questionnaire developed for this study was partially published elsewhere [24] and fully provided in the Supplementary file S1.

Data analysis

Descriptive statistics were used to summarize student characteristics, presenting categorical variables as frequencies (percentages). Continuous variables were presented as means and standard deviations (SD), and medians and interquartile ranges (IQR). Comparisons were made using the Chi-square test or Fisher's exact test

for categorical variables; and T-test and Mann–Whitney U-test for continuous variables.

To identify the factors associated with negative perceptions of educational environment, the perception of educational environment was categorized as a binary outcome: negative perception (i.e., score ≤ 100) and positive perception (i.e., score > 100). We calculated prevalence ratios (PR) using the modified Poisson regression model (i.e., Poisson regression with a robust error variance) with cluster sampling adjustment using the Stata survey commands (svy) to assess the association between potential independent factors and negative perceptions of educational environment [32, 33]. This method was employed to prevent overestimation of association [32, 33], and also used in our previous study [24]. Regression analyses were stratified by sex based on preliminary analyses indicating that the direction and magnitude of several associations differed between male and female students, suggesting effect modification. We fitted 3 seperate Poisson regression models for female and male students. Model 1 was a parsimonious model restricted to year of study, academic ranking, and academic motivation – the three variables with the strongest a priori theoretical justification. Model 2 extended this by including a broader set of variables, which are basic academic information (including academic year, academic ranking, level of English reading skills, failure in at least one clinical exam, failure in at least one theory exam), mental health issues (including stress, depression or anxiety), and self-determined academic motivation. Model 3 further extended Model 2 by including three additional independent factors of perceived quality of clinical learning experience that were not captured by the DREEM tool, including perception that all course objectives were completed, perception that clinical learning was effective relative to time and effort invested, and perception that clinical exam scores accurately reflected students' ability. Smoking and alcohol consumption were excluded from multivariable models due to sparse cell counts in sex-stratified strata, which would have produced unstable estimates. In this population, anxiety disorder had high collinearity with depression; therefore, depression and anxiety were included as a composite variable in the regression analysis as *"Screened positive for depression or anxiety (PHQ 9 ≥ 10 or GAD 7 ≥ 10)"*. In our sensitivity analysis, we re-fitted the full regression models using the continuous DREEM total score as the outcome in linear regression, with covariate structures identical to those in the primary analysis.

Data analysis was performed using STATA 18 (StataCorp LLC, College Station, TX, USA), and plots were generated using R statistical software version 3.4.0. *P*-value of < 0.05 was considered statistically significant.

Ethics

Our research proposal and protocol were approved by the Institutional Review Board (IRB) of Hanoi University of Public Health (No. 280/2015/YTCC-HD3). All invited students were asked for written informed consent after reading the study description. Participants were informed of the minimal risks involved with participation, the confidentiality of their data, and that they could withdraw from the study at any time.

Results

Among 627 students invited to participate in the study, 484 (77.2%) completed the questionnaire. Table 1 presents the characteristics of the study students by sex. The majority of students resided with roommates and reported basic to intermediate proficiency in English reading. Male students failed more clinical exam and theory exam, had significantly lower GPAs and academic rankings (p-values < 0.01), smoked, and consumed more alcohol consumption than female students. Approximately one-fourth of students in both sexes undertook at least two-night shifts per week, and 71% reported having to complete tasks irrelevant to clinical learning objectives during these shifts. The prevalence of self-reported depression was 14.3%, anxiety disorder 7.2%, and stress 51.4%, with no significant sex differences. A high proportion of students (88.0%) reported self-determined academic motivation, which was more common among females ($p = 0.03$). Characteristics of participants by academic year are further shown in Table S1. Additionally, internal consistency was acceptable, with an overall Cronbach's alpha of 0.92 and subscale alphas ranging from 0.63 to 0.83 (Table S2).

Table 2 presents the clinical training experiences and perceptions of clinical learning environment by sex, highlighting similar time allocation across activities but more favorable evaluations among male students. On a clinical study day with scheduled clinical lecture, students spent a median of 180 min on clinical and related academic activities, divided between patient interaction, cases discussion, lectures, and other tasks. A higher proportion of male students reported that course objectives had been fully met (28.9% vs. 18.8%, p = 0.04) and that clinical learning was effective relative to their time and effort (26.0% vs. 15.9%, p = 0.02). Only one-third of both sexes agreed that clinical exam scores accurately reflected students' abilities. Table S3 shows the clinical training experience of students by academic year.

The overall mean DREEM score among medical students was 119.3/200, indicating a perception of the educational environment that was more positive than negative, with no significant sex differences. Most students (87.8%) perceived their educational environment as positive, with 85.5% as "more positive than negative"

Table 1 Characteristics of medical students by sex ($n = 484$)

Characteristics		Male	Female	Total	*P*-value
n (%)		277 (57.23)	207 (42.77)	484 (100)	
Academic year, n (%)	4th year	83 (29.96)	68 (32.85)	151 (31.20)	0.52[a]
	5th year	102 (36.82)	66 (31.88)	168 (34.71)	
	6th year	92 (33.21)	73 (35.27)	165 (34.09)	
Living arrangement, n (%)	Living with family	53 (19.13)	39 (18.84)	92 (19.01)	0.61[a]
	Living alone	33 (11.91)	19 (9.18)	52 (10.74)	
	Living with roommate(s)	191 (68.95)	149 (71.98)	340 (70.25)	
Level of English reading skills, n (%)	Basic	119 (42.96)	98 (47.34)	217 (44.83)	0.59[a]
	Intermediate	132 (47.65)	93 (44.93)	225 (46.49)	
	Proficient	26 (9.39)	16 (7.73)	42 (8.68)	
Failed at least one clinical exam, n (%)		21 (7.58)	6 (2.90)	27 (5.58)	0.03[a]
Failed at least one theory exam, n (%)		170 (61.37)	91 (43.96)	261 (53.93)	< 0.01[a]
GPA of the last academic year, median (IQR)		7.58 (7.21; 7.87)	7.85 (7.54; 8.11)	7.67 (7.34; 7.98)	< 0.01[b]
Academic ranking, n (%)	Excellent (8.0 and above)	49 (17.69)	69 (33.33)	118 (24.38)	< 0.01[c]
	Good (7.0—< 8.0)	189 (68.23)	127 (61.35)	316 (65.29)	
	Average (5.0—< 7.0)	38 (13.72)	11 (5.31)	49 (10.12)	
	Weak (< 5.0)	1 (0.36)	0 (0.00)	1 (0.21)	
Number of night shifts per week, n (%)	< twice a week	212 (76.53)	153 (73.91)	365 (75.41)	0.51[a]
	≥ twice a week	65 (23.47)	54 (26.09)	119 (24.59)	
Had to do irrelevant work in night shifts, n (%)		206 (74.37)	142 (68.60)	348 (71.90)	0.16[a]
Currently smoking, n (%)		8 (2.89)	3 (1.45)	11 (2.27)	0.37[c]
Regular alcohol consumption (> 1 time/month on average), n (%)		100 (36.10)	10 (4.83)	110 (22.73)	< 0.01[a]
Screened positive for depression (PHQ 9 ≥ 10), n (%)		37 (13.36)	32 (15.46)	69 (14.26)	0.51[a]
Screened positive for generalized anxiety disorder (GAD 7 ≥ 10), n (%)		21 (7.58)	14 (6.76)	35 (7.23)	0.73[a]
Stress at medical university, n (%)	No—below median	127 (46.18)	107 (51.94)	234 (48.65)	0.21[a]
	Yes—median and above	148 (53.82)	99 (48.06)	247 (51.35)	
Stress at medical university, median (IQR)		26.00 (22.00; 30.00)	25.00 (22.00; 29.25)	26.00 (22.00; 30.00)	0.71[b]
Academic motivation, n (%)	Non-self-determined	41 (14.80)	17 (8.21)	58 (11.98)	0.03[a]
	Self-determined	236 (85.20)	190 (91.79)	426 (88.02)	

Test of significance comparing the male versus female groups

Abbreviation: *IQR* interquartile range

[a]Chi-square test for categorical variables—displayed as n(%);

[b]Mann-Whitney U test for continuous-skewed variable—displayed as median (IQR);

[c]Fisher-exact test for categorical variables—displayed as n(%);

and 2.3% as "excellent". Among five specific DREEM domains, perception of teachers had the highest mean score, followed by perception of atmosphere, academic self-perception, perception of learning, and social self-perception. We observed no significant sex differences across these five domains (Table 2) but significant differences by adacemic year (Fig. 1). The students' responses to each of the 50 DREEM questions are presented in Figures S1 to S5.

Table 3 shows proportions of negative perception about educational environment by students' characteristics. Proportions of negative perception were significantly higher among students who were in higher academic years, failed at least one clinical exam, assigned with irrelevant tasks during night shifts, smoked, consumed alcohol, experienced depression, anxiety, or stress compared to those without these conditions or behaviors. Meanwhile, students with self-determined academic motivation, those who perceived that course objectives were achieved, that clinical learning was effective, or that exam scores accurately reflected their abilities, showed significantly lower prevalence of negative perception.

Table 4 shows the Poisson regression analysis of potential factors associated with negative perceptions of educational environment, stratified by sex. In all models, consistently in both sexes, students in higher academic years were significantly more likely to report negative perceptions (see further in Table S4 for both sexes). For both models 2 and 3 in males, we found the significant negative association between having self-determined academic motivation (PR: 0.36; 95%CI: 0.23 to 0.56) and negative perception of educational environment. However, depression or anxiety was no longer significant in the full model 3 for male. For females, in the full model

Table 2 Clinical training experiences and perceptions of clinical learning environment by sex

Characteristics		Male	Female	Total	*P*-value
n (%)		277 (57.23)	207 (42.77)	484 (100)	
Time spent in clinical rotations (month), median (IQR)		20 (12; 30)	20 (12; 30)	20 (12; 30)	0.98[a]
Time spent at the hospital in a day with scheduled clinical lecture, median (IQR)	Ask and examine patients (minutes)	60 (30; 60)	60 (50; 90)	60 (32; 90)	0.01[a]
	Discuss clinical cases (minutes)	30 (20; 60)	30 (30; 40)	30 (30; 40)	0.45[a]
	Attend clinician' lectures (minutes)	60 (60; 60)	60 (60; 60)	60 (60; 60)	0.90[a]
	Other activities (e.g., read a book, sit in the hallway, coffee break, nap, etc.) (minutes)	30 (30; 60)	30 (30; 60)	30 (30; 60)	0.01[a]
Time spent at the hospital in a day without scheduled clinical lecture, median (IQR)	Ask and examine patients (minutes)	60 (45; 90)	60 (60; 120)	60 (60; 90)	0.01[a]
	Discuss clinical cases (minutes)	30 (30; 60)	30 (30; 60)	30 (30; 60)	0.17[a]
	Other activities (e.g., read a book, sit in the hallway, coffee break, nap, etc.) (minutes)	60 (30; 90)	60 (30; 60)	60 (30; 90)	0.01[a]
Number of medical records written per week, median (IQR)		1 (1; 2)	2 (1; 2)	1 (1; 2)	0.54[a]
Perceived that all course objectives were completed,[1] n (%)	No	32 (11.55)	30 (14.49)	62 (12.81)	0.04[b]
	Neutral	165 (59.57)	138 (66.67)	303 (62.60)	
	Yes	80 (28.88)	39 (18.84)	119 (24.59)	
Perceived that clinical learning was effective relative to time and effort invested,[2] n (%)	No	43 (15.52)	41 (19.81)	84 (17.36)	0.02[b]
	Neutral	162 (58.48)	133 (64.25)	295 (60.95)	
	Yes	72 (25.99)	33 (15.94)	105 (21.69)	
Perceived that clinical exam scores accurately reflected students' ability,[3] n (%)	No	51 (18.41)	23 (11.11)	74 (15.29)	0.08[b]
	Neutral	128 (46.21)	108 (52.17)	236 (48.76)	
	Yes	98 (35.38)	76 (36.71)	174 (35.95)	
DREEM score (possible range: 0–200)	Median (IQR)	121 (108; 132)	119 (108; 129)	120 (108; 131)	0.35[a]
	Mean (SD)	119.77 (18.73)	118.69 (14.80)	119.31 (17.15)	0.49[c]
DREEM groups, n (%)	Very poor (0–50)	1 (0.36)	0 (0.00)	1 (0.21)	0.01[d]
	Plenty of problems (51—100)	39 (14.08)	19 (9.18)	58 (11.98)	
	More positive than negative (101—150)	227 (81.95)	187 (90.34)	414 (85.54)	
	Excellent (151—200)	10 (3.61)	1 (0.48)	11 (2.27)	
Five specific DREEM domains					
Perception of learning (range: 0–4)	Median (IQR)	2.25 (1.92; 2.58)	2.33 (2.00; 2.58)	2.29 (2.00; 2.58)	0.67[a]
	Mean (SD)	2.27 (0.46)	2.28 (0.35)	2.27 (0.42)	0.71[c]
Perception of teachers (range: 0–4)	Median (IQR)	2.64 (2.36; 3.00)	2.64 (2.45; 2.91)	2.64 (2.36; 2.91)	0.96[a]
	Mean (SD)	2.66 (0.47)	2.66 (0.42)	2.66 (0.45)	0.95[c]
Academic self-perception (range: 0–4)	Median (IQR)	2.38 (2.00; 2.62)	2.25 (2.00; 2.50)	2.25 (2.00; 2.62)	0.11[a]
	Mean (SD)	2.31 (0.51)	2.25 (0.39)	2.29 (0.47)	0.15[c]
Perception of atmosphere (range: 0–4)	Median (IQR)	2.42 (2.08; 2.75)	2.42 (2.08; 2.67)	2.42 (2.08; 2.75)	0.36[a]
	Mean (SD)	2.41 (0.49)	2.38 (0.42)	2.39 (0.46)	0.46[c]
Social self-perception (range: 0–4)	Median (IQR)	2.29 (2.00; 2.57)	2.29 (1.86; 2.57)	2.29 (2.00; 2.57)	0.27[a]
	Mean (SD)	2.27 (0.46)	2.22 (0.45)	2.25 (0.45)	0.21[c]

Test of significance comparing the male versus female groups

Abbreviation: *DREEM* Dundee Ready Educational Environment Measure, *IQR* interquantile range, *SD* standard deviation

[1]Question: You have completed all the course objectives (able to observe diseases, symptoms, and procedures to be observed)?

[2]Question: Do you feel that your clinical learning has been effective relative to the time and effort you put into it?

[3]Question: Do you feel that your clinical exam grades accurately reflect your abilities?

[a]Mann-Whitney U test for continuous-skewed variable—displayed as median (IQR)

[b]Chi-square test for categorical variables—displayed as n(%)

[c]t-test for continuous-normal variable – displayed as mean (SD)

[d]Fisher-exact test for categorical variables—displayed as n(%)

3, those who perceived their clinical learning as effective had approximately 75% lower prevalence of negative perceptions (PR: 0.25; 95%CI: 0.08 to 0.81), whereas males in the same category had about 61% lower (PR: 0.39; 95%CI: 0.21 to 0.74). Additionally, female students who perceived that clinical exam scores accurately reflected students' ability were over half less likely to have negative perceptions. Multivariable linear regression models also showed

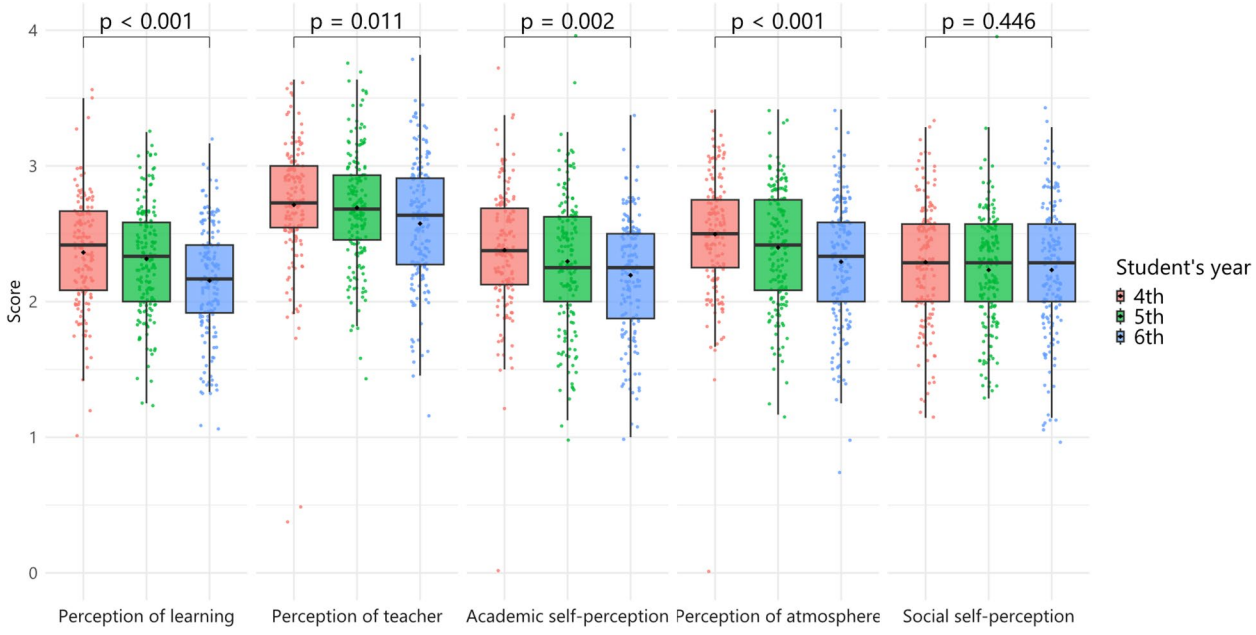


Fig. 1 Boxplot of score across 5 specific DREEM domains by student's academic year

the significant associations between the continuous DREEM scores and academic year, stress, screen-positive depression, academic motivation, fulfillment of learning objectives, perceived effectiveness and assessment fairness of clinical training for both sexes (Table S5, Table S6, Figure S6, and Figure S7).

Discussion

This is the first study to examine medical doctor students' perceptions of the clinical educational environment in Vietnam using the DREEM instrument, and to investigate the factors associated with negative perceptions in a representative sample of the country's largest and most established medical school. Among 484 clinical-year students, 87.8% reported positive overall perceptions of the educational environment (DREEM score > 100/200), with teacher-related domains receiving the highest satisfaction. Senior year status was significantly associated with the negative perception in both sexes, while having self-reported mental health issues (i.e., depression or anxiety and stress) and no academic motivation were significantly associated with negative perceptions among male students only. Additionally, fulfillment of learning objectives, perceived effectiveness and assessment fairness of clinical training were also important associated factors of students' positive perception of the educational environment.

The perceptions of the educational environment during clinical practices among Vietnamese medical students were more positive than negative (119.3 out of 200), reflecting a moderately favorable evaluation. This score is slightly lower than those reported in the Dental Training College in Malaysia (122/200) [18], University of Brasilia in Brazil (123/200) [14], College of Medicine, University of Bahri in Sudan (125.3/200) [15], in 11 dental schools in Korea (125.0/200) [16], some Indian schools (ranged from 123/200 to 136/200) [34–36], UK medical schools, (ranged from 124/200 to 132/200) [17, 18]. However, the DREEM score of our study was higher than that of studies in lower income countries such as Sri Lanka (108/200) [37], Trinidad (109.9/200) [38], Pakistan (107.5/200) [39], and Nigeria (118/200) [18]. Notably, the Iranian study reported a low DREEM score of 99.6/200 [40], reflecting significant dissatisfaction in the educational environment. These variations across countries could be attributed to differences in educational systems, resources, and economic, political, and cultural factors. Vietnam's score reveals a gap that needs to be improved in the educational environment during clinical practice.

Among the five DREEM domains, social self-perception and perception of learning scored lowest in our sample (means 2.25 and 2.27 out of 4.0, respectively), while perception of teachers emerged as a relative strength (2.66/4.0). This pattern aligns with the international DREEM literature, which shows wide cross-national use and recurrent low-scoring items, particularly those reflecting student support and feedback [19, 34, 40]. From a policy standpoint, the lower scores on perception of learning-encompassing how students experience teaching methods, purpose, and engagement-and social self-perception-capturing students' confidence and sense of belonging-are arguably the most actionable [2, 3, 19, 41]. Improvements to curriculum alignment, structured

Table 3 Prevalence of positive and negative perception about educational environment in subgroups of medical students

Characteristics		Positive perception (DREEM score > 100)	Negative perception (DREEM score ≤ 100)	Total	*P*-value
n (%)		425 (87.81)	59 (12.19)	484 (100)	
Sex, n (%)	Male	237 (85.56)	40 (14.44)	277 (100)	0.08[a]
	Female	188 (90.82)	19 (9.18)	207 (100)	
Academic year, n (%)	4th year	144 (95.36)	7 (4.64)	151 (100)	< 0.01[a]
	5th year	149 (88.69)	19 (11.31)	168 (100)	
	6th year	132 (80.00)	33 (20.00)	165 (100)	
Living arragement, n (%)	Living with family	80 (86.96)	12 (13.04)	92 (100)	0.70[a]
	Living alone	44 (84.62)	8 (15.38)	52 (100)	
	Living with roommate(s)	301 (88.53)	39 (11.47)	340 (100)	
Level of English reading skills, n (%)	Basic	188 (86.64)	29 (13.36)	217 (100)	0.56[a]
	Intermediate	198 (88.00)	27 (12.00)	225 (100)	
	Proficient	39 (92.86)	3 (7.14)	42 (100)	
Failed at least one clinical exam, n (%)	Yes	19 (70.37)	8 (29.63)	27 (100)	< 0.01[a]
	No	406 (88.84)	51 (11.16)	457 (100)	
Failed at least one theory exam, n (%)	Yes	225 (86.21)	36 (13.79)	261 (100)	0.24[a]
	No	200 (89.69)	23 (10.31)	223 (100)	
GPA of the last academic year, median (IQR)		7.67 (7.38; 7.98)	7.66 (7.24; 8.00)	7.67 (7.34; 7.98)	0.49[b]
Academic ranking, n (%)	Excellent (8.0 and above)	103 (87.29)	15 (12.71)	118 (100)	0.04[a]
	Good (7.0—< 8.0)	278 (87.97)	38 (12.03)	316 (100)	
	Weak and average (< 7.0)	44 (88.00)	6 (12.00)	50 (!00)	
Number of night shifts per week, n (%)	< twice a week	319 (87.40)	46 (12.60)	365 (100)	0.63[a]
	≥ twice a week	106 (89.08)	13 (10.92)	119 (100)	
Had to do irrelevant work in night shifts, n (%)	Yes	297 (85.34)	51 (14.66)	348 (100)	0.01[a]
	No	128 (94.12)	8 (5.88)	136 (100)	
Currently smoking, n (%)	Yes	7 (63.64)	4 (36.36)	11 (100)	0.03[c]
	No	418 (88.37)	55 (11.63)	473 (100)	
Regular alcohol consumption (> 1 time/month on average), n (%)	Yes	88 (80.00)	22 (20.00)	110 (100)	< 0.01[a]
	No	337 (90.11)	37 (9.89)	374 (100)	
Screened positive for depression (PHQ 9 ≥ 10), n (%)	Yes	50 (72.46)	19 (27.54)	69 (100)	< 0.01[a]
	No	375 (90.36)	40 (9.64)	415 (100)	
Screened positive for generalized anxiety disorder (GAD 7 ≥ 10), n (%)	Yes	25 (71.43)	10 (28.57)	35 (100)	< 0.01[a]
	No	400 (89.09)	49 (10.91)	449 (100)	
Stress at medical university, n (%)	No—below median	219 (93.59)	15 (6.41)	234 (100)	< 0.01[a]
	Yes—median and above	204 (82.59)	43 (17.41)	247 (100)	
Academic motivation, n (%)	Non-self-determined	37 (63.79)	21 (36.21)	58 (100)	< 0.01[a]
	Self-determined	388 (91.08)	38 (8.92)	426 (100)	
Perceived that all course objectives were completed[1], n (%)	No	40 (64.52)	22 (35.48)	62 (100)	< 0.01[a]
	Neutral	270 (89.11)	33 (10.89)	303 (100)	
	Yes	115 (96.64)	4 (3.36)	119 (100)	
Perceived that clinical learning was effective relative to time and effort invested[2], n (%)	No	56 (66.67)	28 (33.33)	84 (100)	< 0.01[a]
	Neutral	268 (90.85)	27 (9.15)	295 (100)	
	Yes	101 (96.19)	4 (3.81)	105 (100)	
Perceived that clinical exam scores accurately reflected student's ability[3], n (%)	No	57 (77.03)	17 (22.97)	74 (100)	< 0.01[a]
	Neutral	205 (86.86)	31 (13.14)	236 (100)	
	Yes	163 (93.68)	11 (6.32)	174 (100)	

Test of significance comparing the positive versus negative perception groups

Abbreviation: IQR interquartile range

[1]Question: You have completed all the course objectives (able to observe diseases, symptoms, and procedures to be observed)?

[2]Question: Do you think that your clinical learning has been effective relative to the time and effort you put into it?

[3]Question: Do you think that your clinical exam grades accurately reflect your abilities?

[a]Chi-square test for categorical variables—displayed as n (%)

[b]Mann-Whitney U test for continuous-skewed variable—displayed as median (IQR);

[c]Fisher-exact test for categorical variables—displayed as n (%)

Table 4 Poisson regression model of negative perception of medical students about the educational environment, stratified by sex

Characteristics	Model 1 for Male (*n* = 275)		Model 1 for Female (*n* = 206)		Model 2 for Male (*n* = 275)		Model 2 for Female (*n* = 206)		Model 3 for Male (*n* = 275)		Model 3 for Female (*n* = 206)	
	PR	95% CI	PR	95% CI	PR	95% CI	PR	95%CI	PR	95% CI	PR	95% CI
Academic year (Ref: 4th year)												
5th year	**2.56**	**1.03–6.34**	1.55	0.42–5.72	2.84	0.99–8.18	1.63	0.24–10.90	**2.44**	**1.00 – 6.00**	1.72	0.25 – 11.92
6th year	**3.34**	**1.40–7.93**	**6.54**	**2.12–20.16**	**3.11**	**1.14–8.47**	**6.40**	**1.11–36.72**	**3.18**	**1.40 – 7.23**	**7.45**	**1.60 – 34.64**
Academic ranking (Ref: Good (7.0- < 8.0))												
Excellent (8.0 and above)	1.49	0.81–2.74	1.24	0.55–2.79	1.92	0.84–4.41	1.26	0.34–4.61	1.99	0.88 – 4.47	0.97	0.17 – 5.57
Weak & Average (< 7.0)	0.61	0.34–1.09	2.83	0.69–11.64	0.56	0.23–1.38	3.59	0.30–43.2	0.44	0.17 – 1.19	3.39	0.60 – 19.27
Self-determined academic motivation (Ref: Non-self-determined)	**0.26**	**0.19–0.35**	0.58	0.28–1.20	**0.35**	**0.25–0.49**	0.77	0.19–3.12	**0.36**	**0.23 – 0.56**	1.70	0.58 – 4.96
Level of English reading skills (Ref: Basic)												
Intermediate and Proficient	–	–	–	–	0.78	0.40–1.54	1.16	0.26–5.19	0.93	0.50 – 1.74	1.54	0.47 – 5.01
Failed at least one clinical exam (Ref: No)	–	–	–	–	2.47	0.87–7.00	2.82	0.27–29.12	2.35	0.96 – 5.74	4.27	0.46 – 39.79
Failed at least one theory exam (Ref: No)	–	–	–	–	0.84	0.39–1.79	0.64	0.30–1.36	0.67	0.32 – 1.39	0.50	0.19 – 1.30
Stress at medical university (Ref: Below median)	–	–	–	–	2.18	0.81–5.90	1.85	0.49–7.00	2.16	0.79 – 5.87	1.40	0.52 – 3.86
Screened positive for depression or anxiety (PHQ 9 ≥ 10 or GAD 7 ≥ 10) (Ref: No)	–	–	–	–	**2.29**	**1.11–4.75**	1.79	0.42–7.58	1.63	0.94 – 2.82	1.65	0.42 – 6.47
≥ 2 night shifts per week (Ref: < twice/week)	–	–	–	–	–	–	–	–	1.27	0.69 – 2.32	2.02	0.57 – 7.15
Perceived that all course objectives were completed[1] (Ref: No)												
Neutral or Yes	–	–	–	–	–	–	–	–	**0.44**	**0.27 – 0.72**	0.42	0.13 – 1.36
Perceived that clinical learning was effective relative to time and effort invested[2] (Ref: No)												
Neutral or Yes	–	–	–	–	–	–	–	–	**0.39**	**0.21 – 0.74**	**0.25**	**0.08 – 0.81**
Perceived that clinical exam scores accurately reflected student's ability[3] (Ref: No)												
Neutral or Yes	–	–	–	–	–	–	–	–	1.26	0.74 – 2.16	**0.48**	**0.23 – 0.98**

Model 1: academic year, academic ranking, and self-determined academic motivation

Model 2: Basic academic information (including academic year, academic ranking, level of English reading skills, failure in at least one clinical exam, failure in at least one theory exam), mental health issues (including stress, depression or anxiety), and self-determined academic motivation)

Model 3: Model 2 + perceived quality of clinical training

Abbreviation: *CI* confidence intervals, *REF* reference level; "–"Not included in the model. Bold text showed significant results.

[1]Question: You have completed all the course objectives (able to observe diseases, symptoms, and procedures to be observed)?

[2]Question: Do you feel that your clinical learning has been effective relative to the time and effort you put into it?

[3]Question: Do you feel that your clinical exam grades accurately reflect your abilities?

peer-learning activities, and student support systems can be achievable through targeted institutional reforms [41].

The DREEM was developed through a multinational Delphi process involving faculty from over 20 countries spanning multiple income settings and has demonstrated acceptable psychometric properties across diverse educational settings [42, 43], including LMIC contexts in Asia and Latin America [1, 19, 25]. The internal consistency observed in our sample is consistent with existing literature [19, 25], lending support to the instrument's applicability in the Vietnamese clinical training context. Prior Vietnamese DREEM studies have been confined to nursing students, consistently reporting high internal consistency (Cronbach's αlpha: 0.94–0.95) [20, 21]. However, the variables that emerged as significant associated factors with negative perception of the learning environment in our study include academic motivation, perceived learning effectiveness, and assessment fairness of clinical training, and mental health status fall out of the scope of the standard DREEM subscales. These factors can serve as unique issues among medical students in low- and middle-income countries (LMICs) settings, where no motivation interview were conducted for university entrance, little educational and well-being support provided may shape the learning experience [25, 44, 45]. We would suggest that future studies in Vietnam, and in comparable LMIC clinical training settings, consider incorporating country-specific educational issues such as self-determined academic motivation, perceived training effectiveness, perceived assessment fairness and support for student wellbeing into the standard DREEM domains to effectively capture Vietnam's unique educational culture for more meaningful insights for interpretation and targeted interventions.

This study found no significant sex differences in students' overall perceptions of the educational environment. These findings are consistent with previous studies at various medical schools in Iran [40, 46], and the United States [47, 48]. However, studies from other countries, including Pakistan [49], Bangladesh [50], Egypt [51], UK [17], and Saudi Arabia [52], found that female medical students tended to perceive the learning environment more positively than male students. These differences suggest that sex-based variations in perceptions of the educational environment may arise in cultural contexts, influenced by the structure of curricula and institutional practices. Despite no significant sex differences in students' perceptions of the environment, the factors were associated with the perception were not the same in each sex group, which could be seen in our regression results. This ensures the inclusion of more qualitative studies aimed at exploring how male and female students internalize specific aspects of the clinical learning environment in Vietnam, thereby informing sex-based strategies to enhance students' satisfaction.

The reversal in the direction of the academic ranking association between sexes warrants careful interpretation. For males, low academic ranking was associated with lower prevalence of negative perception (PR: 0.43; 95% CI: 0.21 to 0.84), whereas for females we found a positive association (PR: 3.62; 95% CI: 1.09 to 12.03). On the one hand, these patterns may reflect statistical suppression effects arising from the small sex-stratified subgroups. On the other hand, they may indicate genuine gender differences in how male and female students perceive the same educational environment. To fully understanding these dynamics would require mixed-methods inquiry; a qualitative interview study exploring sex-differentiated experiences such as clinical supervision, feedback, and social belonging to design targeted, gender-sensitive interventions in clinical training programs.

The positive perception of the educational environment in medical schools tends to decline as students' progress through their academic years. Several studies, including ours, have found that students in later academic years reported more negative perceptions of the educational environment compared to those in earlier academic years [53–55]. Contributing factors to this could include increased stress, work-life balance issues, and changes in student relationships [54, 56]. In the Vietnamese context, 6th-year students also face the added burden of completing their graduation thesis alongside clinical rotations and medical residency entrance examination, which may compound academic stress [5, 24]. This study found that the negative perception of the educational environment was associated with depression among medical students, aligning with findings from previous studies in Malaysia [57] in Peru [58], and Pakistan [59]. Additionally, student's negative perception of the learning environment were also associated with higher level of stress [56], similar to our study results among male students. This indicates that mental health plays a critical role in influencing medical students' perception of learning environment. A negative learning environment can contribute to study-related stressors, including academic pressure and work-life imbalance, thereby exacerbating symptoms of stress and depression. In contrast, students with self-determined academic motivation may exhibit greater resilience, a lower risk of depression [24, 60], and a more positive perception of the learning environment. Therefore, tailored interventions such as motivational interview for university application process, mentorship programs, counseling services, peer-support networks, and strengthening student–teacher relationships may help create a more supportive educational environment, thereby reducing mental health issues and increasing academic motivation.

Our study provided novel findings to the literature that perceived curriculum alignment, clinical training effectiveness and assessment fairness, which was not captured by DREEM, were associated with Vietnamese student's perception of educational environments. This may reflect the inconsistent clinical training and limited support provided to Vietnamese students in clinical settings and the absence of standardized grading guidelines to teachers. Therefore, this finding suggests that medical schools should: (a) pay closer attention to help students accomplish their clinical learning objectives; (b) standardize the clinical assessment rubrics with external moderation to ensure fairness; (c) integrate routine mental health screening into the clinical curriculum with accessible counselling referral pathways; and (d) establish structured faculty mentorship programs, especially for senior clinical students [61–63]. We could consider adding student feedback on these specific aspects into current DREEM domains to get more information, enhance the quality of clinical training, and better support learner outcomes.

This study has some limitations. First, this study was conducted only in the largest Vietnamese medical university, which limits the generalizability of the findings to other medical schools in Vietnam. Second, this is a cross-sectional study so we cannot make causal interpretation on the association between risk factors and outcome; longitudinal studies across multiple universities would be needed to establish temporality and improve generalizability. Thirdly, this study is based solely on the DREEM questionnaire to evaluate the educational environment during clinical practices. Incorporating other tools, such as the Practice-Based Educational Environment Measure (PBEEM), Clinical Learning Environment Scale (CLES), or Postgraduate Hospital Educational Environment Measure (PHEEM), could offer more nuanced perceptions of the clinical educational environment. Fourth, although the DREEM questionnaire underwent translation, back-translation, and pilot testing with 52 students following standard procedures, formal psychometric validation (e.g., confirmatory factor analysis) had not been conducted in Vietnam at the time of data collection. Lastly, the data were collected over a decade ago, prior to major curriculum reforms initiated around 2019–2020, which may not reflect the current student's satisfaction with the educational environment. However, the study can provide a valuable baseline characterization of the clinical educational environment for medical doctor students at Vietnam's largest medical university prior to recent curriculum reforms. This baseline may serve as a reference point for future studies evaluating the impact of curricular changes on students' perceptions.

Conclusion

As the first study to investigate perceptions of the clinical educational environment and mental health status among Vietnamese medical students, this study indicates that students generally viewed the clinical learning environment as more positive than negative (mean score of 119 out of 200). Our results also suggest the targeted gender-based interventions addressing senior-year support, mental health services, mentorship, and assessment fairness can yield measurable improvements in students' perceptions of clinical training.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

LB: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing-original draft, Writing-review & editing; DH: Conceptualization, Formal analysis, Writing-review & editing; TT: Formal analysis, Writing-original draft, Writing-review & editing; MD: Formal analysis, Project administration, Writing-review & editing; TP: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Visualization, Writing-review & editing.

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Data availability

According to our application to the Institutional Review Board of Hanoi University of Public Health, the data cannot be shared publicly because of ethical restrictions to protect the confidentiality of the participants imposed by the IRB. A de-identified dataset is available for researchers who meet the criteria for access to confidential data. Requests for data should be submitted to the Institutional Review Board of Hanoi University of Public Health (irb@huph.edu.vn) and the corresponding author, Dr. Linh Bui (linhbui.leo@gmail.com). Consent for Publication: Not applicable as there are no images or other personal or clinical details of participants are presented that compromise anonymity.

Declarations

Ethics approval and consent to participate

Our research was approved by the Institutional Review Board of Hanoi University of Public Health (No. 280/2015/YTCC-HD3). All invited students were asked for written informed consent after reading the study description and understanding the minimal risks involved with the participation as well as the confidentiality of their data. This study adhered to the Declaration of Helsinki (https://www.wma.net/policies-post/wma-declaration-of-helsinki/).

Consent for publication

Not applicable as there are no images or other personal or clinical details of participants are presented that compromise anonymity.

Competing interests
The authors declare no competing interests.

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